

Akansha Sharma

GenAI Engineer — LLM — Research Engineer
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LinkedIn — GitHub — Hugging Face

Professional Summary

GenAI Engineer with experience in developing multimodal AI systems using LLMs, Vision-Language Models, OCR, and RAG. Skilled in building scalable end-to-end ML pipelines, fine-tuning and deploying AI models, and optimizing inference for production. Passionate about developing accessible, real-world AI solutions.

Technical Skills

AI/ML: Python, PyTorch, TensorFlow, Hugging Face, Transformers, LangChain, Unsloth, LLMs, Generative AI, NLP, OCR, Vision-Language Models, Prompt Engineering, RAG, FAISS, Pinecone
Development: FastAPI, Django, Streamlit, Docker, Git, Linux, AWS, Nginx, CI/CD
Research: AI Accessibility, Model Evaluation, Dataset Curation, Experimental Design
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures, Algorithms, Operating Systems

Professional Experience

AI & STEM Accessibility Research Intern

Feb 2024 – Jul 2024

National Institute of Technology, Durgapur

- Built a multimodal AI pipeline using LangChain, Hugging Face, and PaddleOCR to describe mathematical expressions for visually impaired users in real time
- Developed a Flask-based Chrome extension improving accessibility efficiency by 40%
- Mentored 3 junior members, increasing lab productivity by 25%

Technical Projects

MediLens: Multimodal Healthcare Reasoning System

GitHub

Computer Vision • Qwen2-VL • NLP • Evaluation Framework

- Built a multimodal reasoning pipeline for handwritten prescription understanding and drug interaction analysis
- Implemented context-aware inference for dosage extraction and patient scheduling
- Designed evaluation framework using 500+ real-world samples with structured validation metrics

FormGen-AI: Multimodal Document Intelligence System

FastAPI • Qwen-VL • OCR • RAG

- Engineered a multimodal RAG pipeline for structured information extraction from PDFs and images
- Designed intelligent validation layer for field mapping, error correction, and consistency enforcement
- Optimized deployment lifecycle, reducing release time by 60% via CI/CD automation

Meeting Analyzer: Scalable Audio Intelligence System

Speech-to-Text • NLP • Real-Time Systems

GitHub

- Engineered a real-time multi-speaker pipeline with diarization and topic segmentation
- Applied structured summarization with evaluation loops for improved output consistency

LaTeX Mathematical Reasoning Engine

Transformers • Unsloth • Structured Reasoning

GitHub

- Fine-tuned transformer models on structured LaTeX datasets for symbolic reasoning tasks
- Developed structured parsing framework supporting 200+ mathematical constructs

Education

Asansol Engineering College, West Bengal, India

Expected May 2026

Bachelor of Technology (B.Tech.) in Information Technology

Narayana Junior College, Asansol, India

May 2022

Higher Secondary Education (Science Stream)

Achievements & Publications

- Published the research paper "*Pre-Trained Language Model Augmented with Knowledge (PLMSAWK)*" in the, presenting methods to improve factual reasoning and knowledge integration in Large Language Models.

- Published the conference paper "*A Deterministic Hybrid Agent Architecture for State-Aware Prescription Risk Flagging*" on **NCCCI 2.0 (2026)**, proposing an agent-based framework for intelligent prescription analysis and healthcare risk detection.